

Exercise 5: Date functions

Table: Employees

Q1: Add 6 months to the employee's hire date using

DATEADD ()

emp-id, name, hire-date, hire-plus-6-months

emp-id	name	hire date
1	Alice	2020-01-15
2	Bob	2021-06-10
3	Charlie	2023-03-22

SELECT emp-id, name, hire-date

DATEADD (month, 6, hire-date) AS hire-plus-6-months

FROM employees;

emp-id	name	hire-date	hire-plus-6-months
1	Alice	2020-01-15	2020-07-15
2	Bob	2021-06-10	2021-12-10
3	Charlie	2023-03-22	2023-09-22

Q2: Use DATEDIFF () to find age in days from dob to today.

Student-id, name, age-in-days.

Table: Students

Student-id	name	dob
101	Maya	2025-08-10
102	Ethan	2004-12-01
103	Sienna	2006-03-15

SELECT student_id, name

DATEDIFF (days, dob, current-date) AS age_in_days

FROM students;

student_id	name	age_in_days
101	Maya	65
102	Ethan	7 622
103	Sienna	7 153

Q3. Find how many days are left until each event using

DATEDIFF

= event_id, event_name, days_remaining.

Table 3: Events

event_id	event_name	event_date
1	Seminar	2024-06-15
2	Workshop	2025-09-01
3	Hackathon	2025-01-20

SELECT event_id, event_name

DATEDIFF (DAYS, ^{current-date}~~event-date~~, event-date) AS days_remaining

FROM events;

event_id	event_name	days_remaining
1	Seminar	486
2	Workshop	43
3	Hackathon	267

Q4. Seminar Workshop

Q5. Calculate the number of days between issue-date and due-date
 → invoice-id, issue-date, due-date, days-between

Table 4: Invoices

invoice-id	issue-date	due-date
501	2025-03-10	2025-03-25
502	2025-04-01	2025-04-15
503	2025-04-10	2025-04-20

SELECT invoice-id, issue-date, due-date

DATEDIFF (DAYS, issue-date, due-date) AS day-between
 FROM invoices;

invoice-id	issue-date	due-date	days-between
501	2025-03-10	2025-03-25	13
502	2025-04-01	2025-04-15	12
503	2025-04-10	2025-04-20	8

Q5. Format start-date as 'MONTH YYYY' using TO-CHAR()

→ course-id, name, formatted-date

Table 5: Courses

course-id	name	start-date
201	SQL Basics	2025-05-01
202	Python	2025-06-01

SELECT course-id, name

TO-CHAR (start-date, 'MONTH YYYY') AS formatted-date
 FROM courses;

course-id	name	formatted-date
201	SQL Basics	05 2025
202	Python	06 2025

Q6. Create full date from parts using DATEFROMPARTS()
 PARTS() → built a complete date from separate year, month and day columns.
 → member_id, full_start_date

Table 6: Membership

member_id	start_year	start_month	start_day
1	2023	5	10
2	2022	11	25

SELECT member_id
 DATEFROMPARTS(start_year, start_month, start_day)
 AS full_start_date FROM membership;

member_id	full_start_date
1	2023-05-10
2	2022-11-25

Q7. Extend each renewal date by 1 year using
 DATEADD().

→ sub_id, plan, extended_renewal_date

Table 7: Subscription

sub_id	plan	renewal_date
11	Basic	2025-01-01
12	Premium	2025-03-15

SELECT sub_id, plan
 DATEADD(YEAR, 1, renewal_date)
 FROM subscription

sub_id	plan	extended_renewal_date
11	Basic	2026-01-01
12	Premium	2026-03-15

Q8. Show current date and difference from order-date
USE CURRENT-DATE AND DATEDIFF

→ order-id, order-date, today-date, days-since-order

Table 8: orders

order-id	order-date
1001	2025-04-15
1002	2025-04-10

SELECT order-id, order-date

CURRENT-DATE AS today-date

DATEDIFF (DAY, order-date, CURRENT-DATE) AS

day-since-order

FROM orders;

order-id	order-date	current-date	days-since-order
1001	2025-04-15	2025-10-14	182
1002	2025-04-10	2025-10-14	187

Q9. Extract the year from training-date using DATEPART
OR EXTRACT ()

→ training-id, topic, training-year

Table 9: Training

training-id	topic	training-date
1	Safety	2025-01-10
2	Compliance	2025-02-20

SELECT training-id, topic

EXTRACT (YEAR FROM training-date) AS training-year

FROM Training;

Extract () pulls out part of the date (now we need year)

logging-id	topic	logging-year
1	Safety	2025
2	Compliance	2025

Q10. Extract hour and minute from published-on -

⇒ post-id, title, hour-published, minute-published.

Table 10: Blog-posts.

post-id	title	published-on
1	SQL Tips	2025-04-01 10:15:00
2	Data Cleaning	2025-04-12 16:45:00

SELECT post-id, title ~~published-on~~

EXTRACT (HOUR FROM published on) AS hour-published

EXTRACT (MINUTE FROM published on) AS minute-published

FROM Blog Posts;

post-id	title	hour-published	minute-published
1	SQL Tips	10:00:00	00:15:00
2	Data Cleaning	16:00:00	00:45:00

Q11. Calculate days left until licence expiry using DATEDIFF() and today's date.

⇒ driver-id, licence_expiry, days_left

Table 11: Drivers

driver-id	licence_expiry
301	2025-08-15
302	2028-12-31

SELECT driver-id, licence_expiry
 DATEDIFF (DAYS, CURRENT-DATE, licence_expiry)
 AS days-left
 FROM Drivers;

→ 2025-10-14

driver-id	licence_expiry	days-left
301	2025-08-10	65
302	2023-12-31	67

Q12. Display the current timestamp and calculate seconds since the message was sent.

→ message-id, sent_timestamp, current_timestamp, seconds-since-sent.

Table 12: Messages

Message-id	sent_timestamp
1	2025-04-19 09:32:45
2	2025-04-18 23:59:59

SELECT message-id, sent_timestamp
 CURRENT_TIMESTAMP AS current_timestamp,
 DATEDIFF (SECOND, sent_timestamp, current_timestamp) AS
 seconds-since-sent

FROM Messages;

Message-id	sent_timestamp	current_timestamp	seconds-since-sent
1	2025-04-19 09:32:45		
2	2025-04-18 23:59:59		

Q13. Add 15 days to return-date using DATEADD ()
to show restock-date

⇒ return-id, return-date, restock-date

Table 13: Returns

return-id	return-date
901	2025-04-05
902	2025-04-01

SELECT return-id, return-date ^{adds 15 days}
DATEADD (DAYS, 15, return-date) AS restock-date
FROM returns;

return-id	return-date	restock-date
901	2025-04-05	2025-04-20
902	2025-04-01	2025-04-16

Q14. Convert assigned-on to date using TO-DATE ()

⇒ assign-id, assigned-on-date

Table 14: Assignments

assign-id	assigned-on
1	2025-03-01
2	2025-03-05

SELECT assign-id ^{# convert a string value to an actual date format.}
TO-DATE (assigned-on, 'YYYY-MM-DD') AS assigned-on-
date

FROM assignments.

assign-id	assigned-on-date
1	2025-03-01
2	2025-03-05

Q15. Convert scheduled - time to formatted string like
'April 19, 2025 at 2:00 PM' using TO_CHAR()

⇒ meeting-id, formatted-meeting-time

Converted date/

time to a formatted string

Table 15: Meetings

meeting-id	scheduled-time
1	2025-04-19 14:00:00
2	2025-04-19 09:30:00

'April, 19, 2025
at 2 PM'

SELECT meeting-id

TO_CHAR(scheduled-time, 'Month DD, YYYY "at"'

HH:MM AM') AS formatted-meeting-time

FROM Meetings;

meeting-id	formatted-meeting-time
1	April 19, 2025 at 14:00 PM
2	April 19, 2025 at 9:30 AM